3.2.16

EE24BTECH11065 - spoorthi yellamanchali

Question:
The construction of triangle ABC, given that $BC = 6cm$, $\angle C = 60°$ is possible when difference of AB and AC is equal to

1) $3.2cm$
2) $3.1cm$
3) $3cm$
4) $2.8cm$

Solution:

Variable	Description
B	vertex on side of given length of the triangle
C	vertex on side of given length of the triangle
A	third vertex of the triangle

TABLE 4: Variables Used

1) $BC = 6cm$
2) $\angle C = 60°$
3) let $d = |AB - AC|$

The Law of Cosines is given by:

$$AB^2 = BC^2 + AC^2 - 2 \cdot BC \cdot AC \cdot \cos(\angle C)$$

Given $BC = 6$cm and $\angle C = 60°$, we know:

$$\cos(60°) = 0.5$$

Then ,

$$6AC = 36 - AC^2 - AB^2$$

Let the difference between AB and AC be d,
then, there are two possibilities,

$$1)AB = AC + d;$$

$$2)AC = AB + d$$

1) let $AB = AC + d$.
Substitute these values into the Law of Cosines equation:

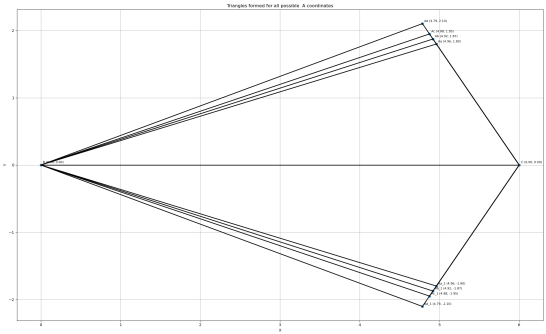
$$6AC = 36 - AC^2 - (AC + d)^2$$

$$AC = \frac{36 - d^2}{6 + 2d}$$

for the first case, For AC to be positive, $d < 6cm$. all the options satisfy the following condition. therefore,d can take the values $3.2cm$,$3.1cm$,$3cm$,$2.8cm$.

2) let $AB = AC - d$ substitute these values in the law of cosines equation:

$$6AC = 36 - AC^2 - (AC - d)^2$$

$$AC = \frac{36 - d^2}{6 - 2d}$$

for the second case,
For AC to be positive, $d < 3cm$. only one option satisfies this condition. therefore,d can only take the value $2.8cm$. For each of the d value there are different coordinates of A.